Clinical, metabolic, and endocrine parameters in response to metformin and lifestyle intervention in women with polycystic ovary syndrome: a random...

The aim of this study was to evaluate the effects of metformin in addition to diet and exercise on endocrine and metabolic disturbances in women with polycystic ovary syndrome (PCOS) in a prospective, double-blind, randomized, placebo (PBO) control trial. Thirty women with insulin resistance and PCOS received lifestyle modification and 1500 mg of metformin or placebo for 4 months. Before and after treatment, body mass index, waist/hip ratio, blood pressure, hirsutism, and menstrual patterns were evaluated. Serum concentrations of gonadotropins, androgens, progesterone, glucose, insulin, and lipids were measured. Lifestyle interventions resulted in similar weight and menstrual cycle's improvements in both groups. A significant reduction in serum fasting insulin, HOMA index, waist and testosterone levels was only observed with metformin. There were no significant changes in androstenedione, dehydroepiandrosterone sulfate, gonadotropins, and lipids levels. No other changes were observed in hirsutism or blood pressure. These findings suggest that metformin has an additive effect to diet and exercise to improve parameters of hyperandrogenism and insulin resistance. Although, a small decrease in body weight trough lifestyle changes could be enough to improve menstrual cycles in insulin-resistant women with PCOS.